

Qatar-1b

R-2437

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_111006 · created 2026-08-22T01:29:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455840.75698 +/- 0.00037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.016
Transit depth (Rp/Rs)^2	2.84 +/- 0.53 [%]
Orbital Inclination (inc)	83.42 +/- 1.16
Airmass coefficient 1 (a1)	1.206 +/- 0.018
Airmass coefficient 2 (a2)	-0.1227 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	3.21
Optimal Annulus	12.36
Transit Duration (day)	0.0666 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.168 ± 0.016 vs 0.1463 (deviation 1.31σ)
Duration fit vs published	0.0666 d vs 0.0692 d (deviation 0.34σ)
Mid-time O–C	0.9 min
Depth SNR	10.2
Residual scatter	1.53 %

Light curve

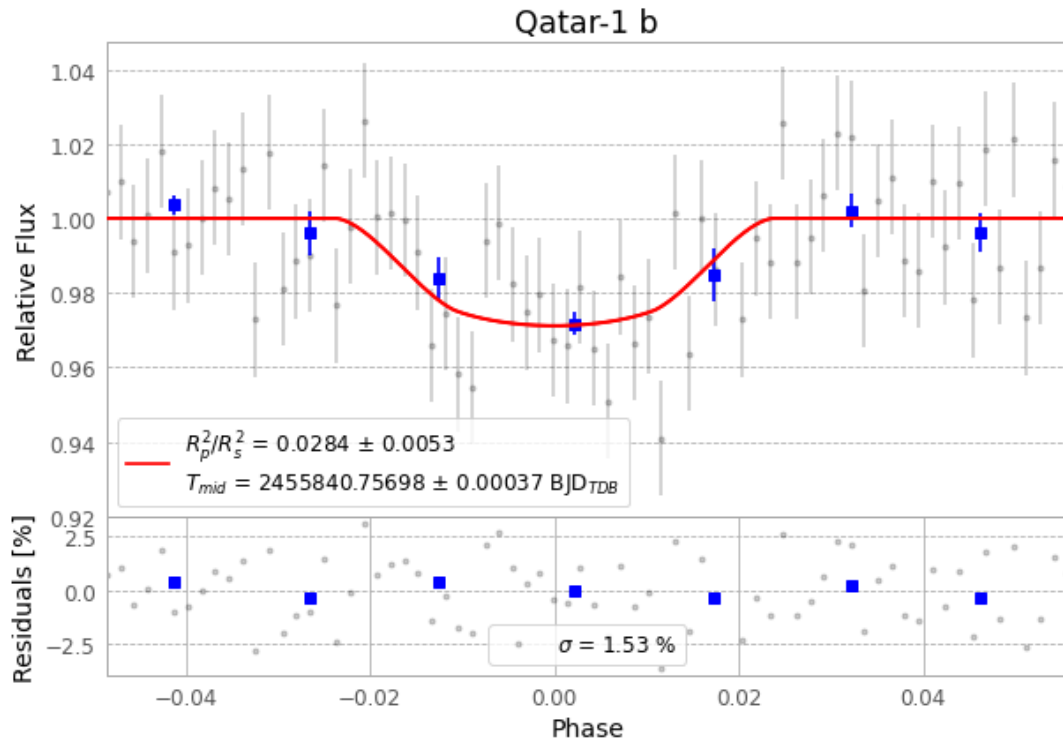


Figure 1 — FinalLightCurve_Qatar-1 b_2011-10-05.png (SHA-256 a197ead7719d7b46...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 330], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6f05d9a4f01b2791...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1111006042712.FITS → Qatar-1111006080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-111006.log (55543c431b2a...), FinalLightCurve_Qatar-1 b_2011-10-05.png (a197ead7719d...), FinalLightCurve_Qatar-1 b_2011-10-05.csv (48c2bba5a4c2...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 01:29Z.